29. Speed, Distance & Time

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Finding speed:

A cyclist travels 24 km in 1 hour. What is her speed?

Answer: Speed = distance ÷ time = 24 ÷ 1 = **24 km/h**.

Example 2 — Finding distance:

A car travels at 25 km/h for 4 hours. How far does it travel?

Answer: Distance = speed × time = 25 × 4 = **100 km**.

Example 3 — Finding average speed:

A train covers 160 km in 4 hours. What is its average speed?

Answer: Speed = distance ÷ time = 160 ÷ 4 = **40 km/h**.

Example 4 — Finding time:

A car travels at a constant 50 km/h to cover 200 km. How long does the journey take?

Answer: Time = distance ÷ speed = 200 ÷ 50 = **4 hours**.

Example 5 — Multi-step: average speed over two legs:

A car travels 80 km in the first 2 hours, then 40 km in the next 2 hours. What is its average speed for the whole journey?

Answer: Total distance: 80 + 40 = 120 km. Total time: 2 + 2 = 4 hours. Average speed = 120 ÷ 4 = **30 km/h**.

Foundation

15 questions

1. A car travels 50 km in 1 hour. What is its speed?

2. Which formula do we use to find distance travelled? A) Speed × Time B) Speed ÷ Time C) Speed − Time D) Time ÷ Speed

3. A cyclist rides at 18 km/h for 4 hours. How far does she travel?

4. A plane flies 300 km in 2 hours. What is its speed?

5. Which unit correctly measures speed? A) km B) hours C) km/h D) kg

6. A walker moves at 4 km/h for 5 hours. How far does she walk?

7. A car travels 80 km in 2 hours. What is its average speed?

8. A train travels at 70 km/h for 3 hours. How far does it travel?

9. Select ALL of the speed calculations below that are correct: A) 80 km in 2 hours = 40 km/h B) 50 km in 5 hours = 10 km/h C) 60 km in 2 hours = 20 km/h D) 36 km in 4 hours = 9 km/h E) 20 km in 2 hours = 5 km/h

10. A car travels 120 km in 3 hours. What is its speed?

11. A runner covers 10 km in 2 hours. What is her speed?

12. Which of these speeds is the fastest? A) 20 km/h B) 55 km/h C) 35 km/h D) 45 km/h

13. A coach travels at 40 km/h for 3 hours. How far does it travel?

14. A car takes 5 hours to travel 250 km. What is its average speed?

15. A cyclist travels 10 km in the first hour and 10 km in the second hour. What is his average speed over the 2 hours?

Standard

20 questions

1. A ship sails 200 km in 4 hours. What is its speed?

2. A tractor drives at 18 km/h for 4 hours. How far does it travel?

3. A car travels 210 km in 3 hours. What is its speed? A) 60 km/h B) 70 km/h C) 80 km/h D) 90 km/h

4. A horse trots at 15 km/h for 3 hours. How far does it travel?

5. A train travels 240 km in 3 hours. What is its speed?

6. A bus travels at 35 km/h for 5 hours. How far does it travel?

7. A plane flies at 500 km/h for 3 hours. How far does it travel? A) 1,000 km B) 1,200 km C) 1,500 km D) 1,800 km

8. A cyclist covers 36 km in 3 hours. What is her speed?

9. A car travels 280 km in 4 hours. What is its speed?

10. A walker travels at 7 km/h for 4 hours. How far does he walk?

11. Select ALL statements that are TRUE about a car travelling at 60 km/h: A) In 2 hours it travels 120 km B) In 3 hours it travels 200 km C) In 1 hour it travels 60 km D) In 4 hours it travels 240 km E) In 5 hours it travels 300 km

12. A train takes 5 hours to travel 300 km. What is its average speed?

13. A coach travels at 35 km/h for 4 hours. How far does it travel?

14. Which journey has the greatest speed? A) 150 km in 3 hours B) 120 km in 2 hours C) 160 km in 4 hours D) 100 km in 2 hours

15. A car travels 120 km in the first 3 hours and 120 km in the next 3 hours. What is its average speed for the whole journey?

16. A ship sails 160 km in 4 hours. What is its speed?

17. A bus travels at 25 km/h for 6 hours. How far does it travel?

18. A train travels 330 km in 3 hours. What is its speed?

19. A car travels 100 km in the first 2 hours and 100 km in the next 2 hours. What is its average speed for the whole journey?

20. Which journey has the greatest speed? A) 90 km in 3 hours B) 140 km in 4 hours C) 80 km in 2 hours D) 150 km in 5 hours

Hard

5 questions

1. On a long drive, a family covers 80 km during the first 3 hours, then a further 40 km over the next 2 hours. Work out their average speed for the whole drive.

2. Travelling at a steady 60 km/h, a train completes a 150 km route. How many hours does the trip take? A) 2 hours B) 2.5 hours C) 3 hours D) 3.5 hours

3. Maya cycles 15 km over the first 2 hours of a trip, then covers another 15 km in the following hour. What is her average speed across the whole trip?

4. Flying at a constant 250 km/h, an aeroplane needs to cover 750 km. How many hours will the flight last?

5. A delivery van drives at 30 km/h for the first 2 hours of its route, then speeds up to 40 km/h for 2 more hours. What total distance has it covered?

© PassFrame Tuition — for registered student use